

First-Principles-Based Kinetic Monte Carlo Model of Hydrogen Evolution Reaction under Realistic Conditions: Solvent, Hydrogen Coverage and Electric Field Effects

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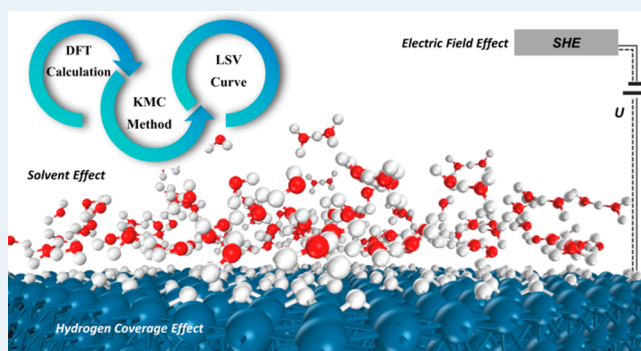


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ABSTRACT: The hydrogen evolution reaction (HER) plays an important role in electrocatalytic water splitting. Despite the progress on the development of HER catalysts, the dynamic evolution of HER reaction under realistic electrochemical conditions considering the electric field, solvent, and hydrogen coverage effects is still unclear. In this study, a first-principles-based H surface coverage and potential-dependent kinetic Monte Carlo (KMC) HER model on the Pt (111)/Pt (100) surface is presented. The reaction kinetics and electronic structure analysis of HER on Pt surfaces in the presence of dihydrated proton (H_3O_2^+) and H surface coverage is investigated using density functional theory (DFT). The HER KMC model was developed based on the DFT-calculated energetics. The KMC simulation results showed that consideration of H_3O_2^+ species and dynamic evolution of H coverage is essential for accurate description of HER reaction on the Pt catalyst, which fits well with HER polarization data. Moreover, sensitivity analysis shows that HER on Pt (111) is mainly affected by the Tafel step. On the Pt(100) surface, HER is primarily governed by the Heyrovsky pathway. Surface species evolution analysis demonstrates that the high working potential accelerated the formation of $[\text{Pt-2H}]$ species, leading to increased H coverage and accelerating the HER process. The predicted weakened H binding strength and increased H coverage at high HER working potential was verified by *in situ* attenuated total reflection Fourier transformed infrared spectroscopy analysis. Overall, the proposed DFT-KMC model represents the state-of-art dynamic simulation of catalytic HER reaction, providing important insights into the evolution of HER under realistic operation conditions.



KEYWORDS: hydrogen evolution reaction, kinetic Monte Carlo, solvent effect, hydrogen coverage, electric field

1. INTRODUCTION

Hydrogen is an important green energy carrier due to its high energy density and zero-emission of carbon dioxide.¹ Hydrogen production from electrocatalytic water splitting is regarded as a promising and sustainable approach.^{2,3} The hydrogen evolution reaction (HER) is the cathodic half-reaction in water splitting.⁴ During the last decades, extensive studies have been devoted to the development of efficient HER catalysts toward high energy efficiency water splitting reaction. However, the successful fabrication of active HER catalysts relies on the thorough understanding of the HER mechanism on the catalyst surface.

The catalytic HER reaction is known to follow two different mechanisms, namely, the Volmer–Heyrovsky and Volmer–Tafel pathways.⁵ The evaluation of catalytic HER activity is usually performed using Gibbs free energy-based equilibrium potential analysis using density functional theory (DFT) calculations.^{6,7} To better approximate realistic HER conditions, DFT studies focused on the effects of solvent,⁸ hydrogen coverage⁹ and electric field¹⁰ on HER activity have

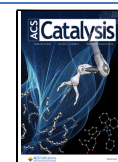
been reported. However, those thermodynamic-based DFT studies neglect crucial reaction kinetic information, such as activation barriers and surface active site evolution. Accordingly, those reports can only consider an “educated-guess” in predicting the activity and rate-determining steps on electrocatalysts.¹¹ More importantly, the atomic level catalytic reaction information obtained by DFT calculation is “static”. In fact, during the course of HER reaction, the reaction pathways and energetics change as a function of the catalyst surface state and the working potential. Therefore, a time-dependent kinetic model that is capable of reflecting the dynamic evolution of HER and catalyst surface state under

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critical realistic reaction conditions is essential to better understand the key HER mechanisms.

Kinetic Monte Carlo (KMC) is an effective method to capture the transient evolution of the reactions taking place on catalytic surface.^{12,13} KMC simulations can bridge the elementary reactions and macroscopic experimental performance of the catalyst under realistic conditions.^{14,15} For example, Kim et al.¹⁶ developed a KMC model to study the role of sulfur vacancy in MoS₂ catalyzed HER reaction by simulating the HER polarization curve. In that study, hydronium ions (H₃O⁺) are considered for the protons, and the effect of the operation voltage was also investigated. That study also reported that the protons in the solvent might be better represented by H₅O₂⁺ or H₉O₄⁺ species rather than H₃O⁺.¹⁷ In another study, Pašti et al.¹⁸ constructed a KMC model considering the H coverage effect to investigate the role of H spillover, migration of activated H atoms onto a reducible metal oxide support, which can enhance the HER performance. However, this HER KMC model, the effects of the electric field and solvent effects are not considered. Also, the H coverage considered in that study is within 1 monolayer (ML). It has been reported that the H surface coverage in HER can reach 2 ML in acidic electrolytes rather than 1 ML.^{19–21} The high H coverage claim is also supported by the reported HER Pourbaix diagram.²² Therefore, it is expected that an accurate KMC HER model should take the electric field, reasonable dihydrated proton solvent model, and dynamic H coverage effect into consideration, which is essential for the realistic modeling of the catalytic HER reaction.

In this study, a DFT-based KMC model is developed to predict the transient evolution of HER reaction on Pt catalysts considering realistic electrochemical conditions. In the DFT calculations, the energy barrier of each HER elementary reaction on Pt(111) and Pt (100) in the presence of H₅O₂⁺ with different H surface coverage is estimated. The crystal orbital Hamilton population (COHP), d-band-center, work function, and average H atom charge were also calculated to analyze the activity origin of Pt catalysts. Based on the DFT-calculated energetics, a KMC model was developed, which enables self-adjusting and tracking of the H surface coverage on Pt (111) and Pt (100) in HER. HER polarization curves on Pt (111) and Pt (100) were predicted using the DFT-KMC model. The accuracy of the DFT-KMC model was validated by comparing its predictions with experimental linear sweep voltammetry (LSV) data. The results show that HER on Pt (111) is mainly controlled by the Tafel pathway, and HER is governed by the Heyrovsky pathway on Pt (100). The predicted HER catalytic surface behavior is also verified by *in situ* attenuated total reflection Fourier transformed infrared spectroscopy (ATR-FTIR) analysis. This study highlights the significance of considering H coverage and H₅O₂⁺ ions in the theoretical analysis of HER reaction. The proposed DFT-KMC model represents a state-of-art dynamic HER model that provides new insights on the transient evolution of the electrochemical reaction mechanism under realistic operation conditions.

2. COMPUTATIONAL AND EXPERIMENTAL METHODS

2.1. Catalyst Models. In this study, the best-performing HER electrocatalyst Pt is selected for the catalytic modeling of HER reaction. Two typical Pt catalyst surfaces (111) and (100) planes were selected.^{23–25} The bulk Pt lattice constant is

calculated to be 3.92 Å, which is in good agreement with the experimental data.^{23,26} The Pt (111) and Pt (100) slab models are constructed using a (3 × 3) slab with four atomic layers. During the structural optimization, the positions of the atoms in the top three atomic layers were allowed to relax, whereas the bottom layer is fixed at their optimized bulk position. To avoid interactions between periodic Pt slab, a 15 Å vacuum thickness was employed.^{27,28} Hydrogen adsorption on different surface sites, including the top, bridge, fcc-hollow, and hcp-hollow, have been considered (Table S1). The top and bridge sites adsorption configuration were calculated as most stable under high hydrogen coverage, which is consistent with the previous studies.^{29–31}

2.2. DFT Calculations and Workflow. The DFT calculations were performed using the Vienna ab initial Simulation Package (VASP) implemented with Blochl's all-electron-like projector augmented wave (PAW) method.^{32,33} The Perdew–Burke–Ernzerhof functional is adopted.³⁴ The spin polarization is considered in the calculations.³⁵ A Monkhorst–Pack 2 × 2 × 1 k-point mesh is applied for integrating the Brillouin zone.³⁶ The cutoff energy is set to 400 eV.³⁷ A convergence test was conducted for the k-point mesh and the energy cutoff (Figure S1). Climbing image nudged elastic band (CI-NEB) method was applied to determine the transition states (TS) in the reaction pathways.³⁸ The numbers of intermediate images used in NEB calculations are determined based on the distance between the initial and final states, which are usually set as 4 or 6. The activation energy (ΔE_a) of HER reaction step is calculated as follows:

$$\Delta E_a = E_{TS} - E_{INI} \quad (1)$$

where E_{TS} represents the energy of transition state, and E_{INI} represents the energy of the initial state of the reaction.²⁵

The adsorption energy (E_{ads}) is calculated according to the following equation:

$$E_{ads} = E_{A/slab} - E_A - E_{slab} \quad (2)$$

where $E_{A/slab}$ represents the total energy of the optimized adsorption system, and E_A and E_{slab} are the energies for the isolated reaction intermediate A and the clean slab, respectively. For the protonation steps in HER, the proton is represented by H₅O₂⁺. This is because H₅O₂⁺ is more reasonable representing the hydronium ions than H₃O⁺.³⁹

Figure 1 provides an workflow overview of this study. In order to better simulate the HER reaction process under realistic conditions, several crucial factors, including H surface coverage, working potential, and solvation effects, were considered. First, all possible elementary reactions in the HER process were determined, and their energy barriers were calculated by DFT. Then, the prefactors and rate constants were obtained through the transition state theory, which was used to construct the kinetic database required for the KMC model.^{40,41} Subsequently, the dynamic HER process was simulated using the proposed KMC model, in which the time evolution of the surface species on Pt catalysts was tracked. Accordingly, the HER reaction mechanism under realistic (transient) working conditions was identified. The DFT-KMC model establishes a connection between microscopic (atomic) electronic state properties and experimentally measured HER performance. The results of the KMC simulation can be represented as HER LSV curves, surface species evolution, and surface coverage variation. This allows for a direct comparison

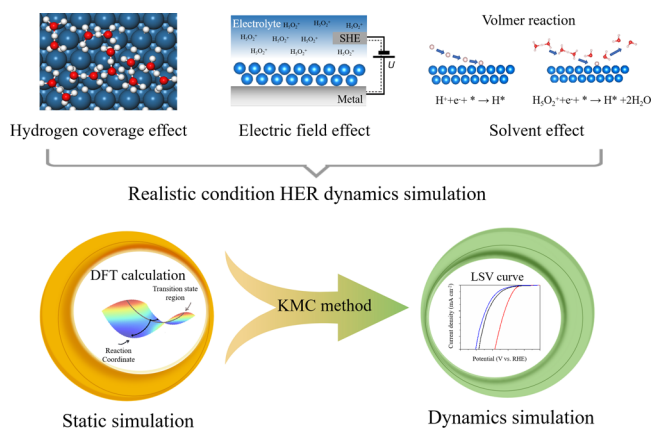


Figure 1. Schematic illustration of the DFT-KMC framework of HER modeling.

and validation of the proposed HER mechanism under realistic working conditions with experimental data.

2.3. Intrinsic Kinetic Calculations and Possible HER Reaction Cycles. The kinetic parameters used in the construction of the first-principles-based KMC model were obtained from DFT calculations. Transition state theory was used to calculate the rate constants (unit: s^{-1}) for the nonelectrochemical surface reaction in HER.⁴¹

$$k = A \times \exp\left(-\frac{E_a}{k_B T}\right) \quad (3)$$

$$A = \frac{k_B T}{h} \frac{q_{\text{vib}}^{\text{TS}}}{q_{\text{vib}}^{\text{R}}} \quad (4)$$

where A is the prefactors (s^{-1}); E_a is the energy barrier of the reaction; k_B represent the Boltzmann's constant; T is the absolute temperature (K); h is the Planck's constant; $q_{\text{vib}}^{\text{TS}}$ and $q_{\text{vib}}^{\text{R}}$ denote the vibrational partition functions of the transition

state configuration and reactants, respectively. Moreover, the vibrational partition function was calculated as follows:

$$q_{\text{vib}} = \prod_{i=1}^n \frac{e^{-h\nu_i/2k_B T}}{1 - e^{-h\nu_i/k_B T}} \quad (5)$$

where ν_i is the frequency of the i^{th} vibrational mode (cm^{-1}).

For the HER charge transfer elementary steps, the forward and backward reaction rates are calculated as follows:⁴²

$$k_{\text{for}} = \frac{k_B T}{h} \frac{q_{\text{vib}}^{\text{TS}}}{q_{\text{vib}}^{\text{R}}} \exp\left(-\frac{(E_a - \beta n_i e \eta)}{k_B T}\right) \quad (6)$$

$$k_{\text{back}} = \frac{k_B T}{h} \frac{q_{\text{vib}}^{\text{TS}}}{q_{\text{vib}}^{\text{R}}} \exp\left(-\frac{(E_a + (1 - \beta) n_i e \eta)}{k_B T}\right) \quad (7)$$

where n_i represents the number of transferred electrons in the charge transfer step, e is the charge of an electron (equivalent to 1.6×10^{-19} C), β is the symmetry factor (which is set to 0.5), and η is the overpotential.⁴³ One can expect that, for a protonation reaction, the higher working potential would accelerate the reaction. For a deprotonation reaction, the higher working potential would hinder the reaction.

The reported HER pathways consist of three elementary reaction steps: Volmer reaction, Heyrovsky reaction, and Tafel reaction. The Volmer and Heyrovsky are charge transfer reaction steps, and thus, their reaction kinetics would be affected by the working potential. The Tafel reaction is a nonelectrochemical step, which is independent of working potential.¹⁶ It is known that the HER elementary reactions can be affected by the H surface coverage.^{44,45} As described below, one active Pt site might accommodate two H^* at the same time under high H surface coverage (>1 ML). Figure 2 shows the four possible HER reaction cycles at low and high H surface coverage. For PATH I, under low H coverage condition, the Pt site accepts one electron forming a $[\text{Pt}]^-$ site, then reacts with H_5O_2^+ to form $[\text{Pt-H}]$. The $[\text{Pt-H}]$ site is then charged again, forming the $[\text{Pt-H}]^-$ site. Subsequently, the

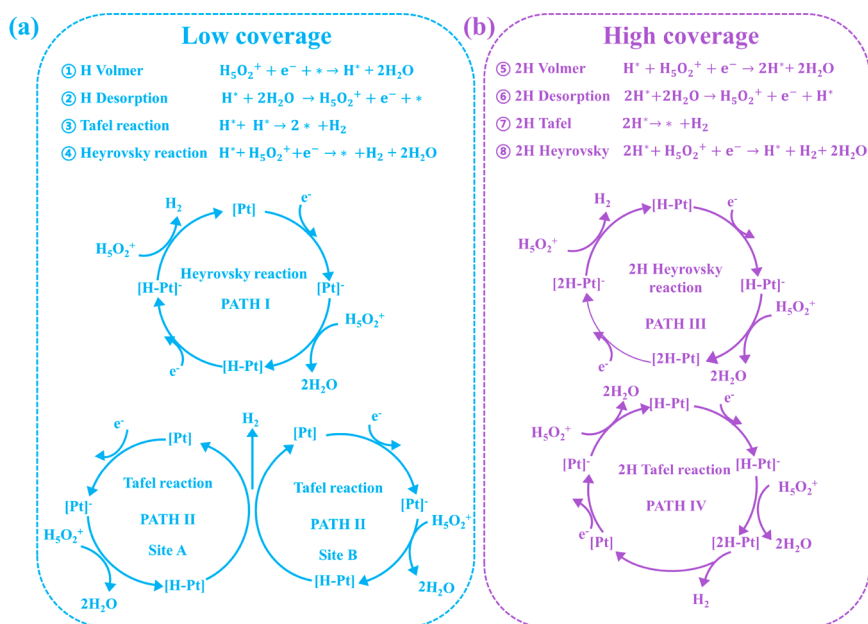


Figure 2. Possible HER reaction cycles at low (a) and high (b) H coverage condition.

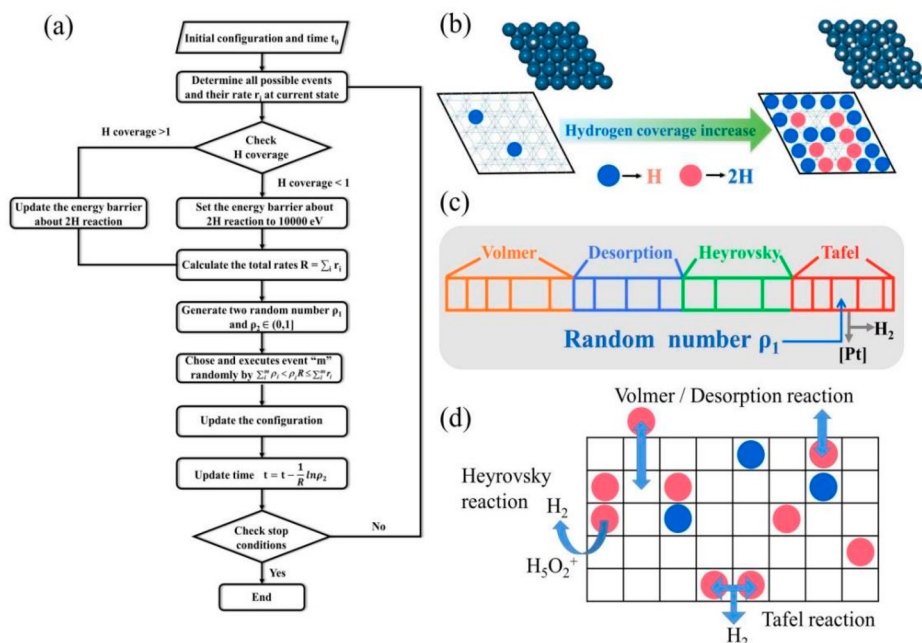


Figure 3. (a) KMC algorithm used to simulate the HER reaction. (b) Demonstration of the KMC simulation of HER on the catalyst lattice. (c) Schematic of the HER event selection process. (d) Schematic of the KMC lattice.

adsorbed H^* directly reacts with H_3O_2^+ to produce H_2 , releasing a clean Pt site and completing the HER cycle. This process is referred to as the Heyrovsky reaction pathway. Similarly, when it comes to the Tafel pathway, after the formation of $[\text{Pt-H}]$, two adjacent $[\text{Pt-H}]$ sites react with each other to produce H_2 , releasing two clean Pt sites. That is, the PATH II pathway involves two Pt sites. With the increase of H surface coverage, one Pt site might be occupied by two H^* , leading to the formation of $[\text{Pt-2H}]$ state. Accordingly, the H^* participated reaction at $[\text{Pt-2H}]$ would undergo different energetics with that of $[\text{Pt-H}]$. Therefore, the HER reactions occurs at $[\text{Pt-2H}]$ sites would be considered as different reactions in the KMC model. For example, in the KMC simulation, there would be two types of Volmer reactions: (i) the proton in H_3O_2^+ ion adsorbed on one surface Pt site, forming a $[\text{Pt-H}]$ state; (ii) the proton in H_3O_2^+ ion assigned to Pt site that already absorbed one hydrogen atom and then forms a $[\text{Pt-2H}]$ state. The energy barrier of these two cases is different. That is, the effect of H–H interactions on the kinetics of Volmer reaction is somehow considered in the KMC simulation.

Also, one can expect that, upon charging to $[\text{Pt-2H}]$, the H_3O_2^+ ion reacts with this site to produce H_2 , forming the $[\text{Pt-H}]$ site (PATH III). If the 2H^* on $[\text{Pt-2H}]$ reacts with each other, the reaction would follow the 2H Tafel pathway (PATH IV).

2.4. KMC Framework. This study presents a coverage and potential dependent KMC model for HER reaction. The present model allows for the self-adjustment of H coverage during the reaction process with updated kinetic parameters. Tables S2–S5 in the Supporting Information shows the event list and kinetic parameter used in this KMC simulation of all the elementary reactions under different H coverage. Moreover, Figure 3a shows the KMC framework, which involves the following steps:

- (i) Define the simulation lattice with an initial surface configuration and initial time t_0 . In this study, an empty

$N \times N$ two-dimensional lattice represents clean Pt (111) or Pt (100) surface was set as the initial surface configuration. That is, all the positions in the lattice are considered active Pt sites. Figure 3b illustrates how the HER reaction evolves on the KMC latticed model.

- (ii) Enumerate all possible events that might occur on the current lattice configuration, and estimate the rate of each event at this specific surface configuration.
- (iii) Calculate the total rate R by adding the reaction rates r_i for all the possible i events considered, and generate a pair of random uniformly distributed numbers ρ_1 and ρ_2 in the range of $(0,1]$.
- (iv) The total rate R is used along with the random number ρ_1 to select an event (i.e., a reaction i) from all the events considered in step (ii). The event is chosen by comparing the cumulative reaction probability distribution P_i given below against ρ_1 :

$$P_i = \sum r_i / R \quad (8)$$

where r_i and R were in step (iii). If the random value assigned to ρ_1 is between P_{i-1} and P_i , the event i will then be selected as the executed event in catalyst surface (Figure 3c).

- (v) Implement the selected event, and update the lattice configuration and event table list, which include all the possible events might occur on the updated lattice configuration. For instance, if one H adsorption event is implemented, the empty lattice site would be occupied by one H^* atom. Meanwhile, since this site has been occupied by H^* atom, H adsorption event on the site is removed. Also, because the H^* atom might react with the H_3O_2^+ , H_2 production event as possible reaction will be added into the possible event list (Figure 3d).
- (vi) Calculating the time interval for the executed surface event by the equation:

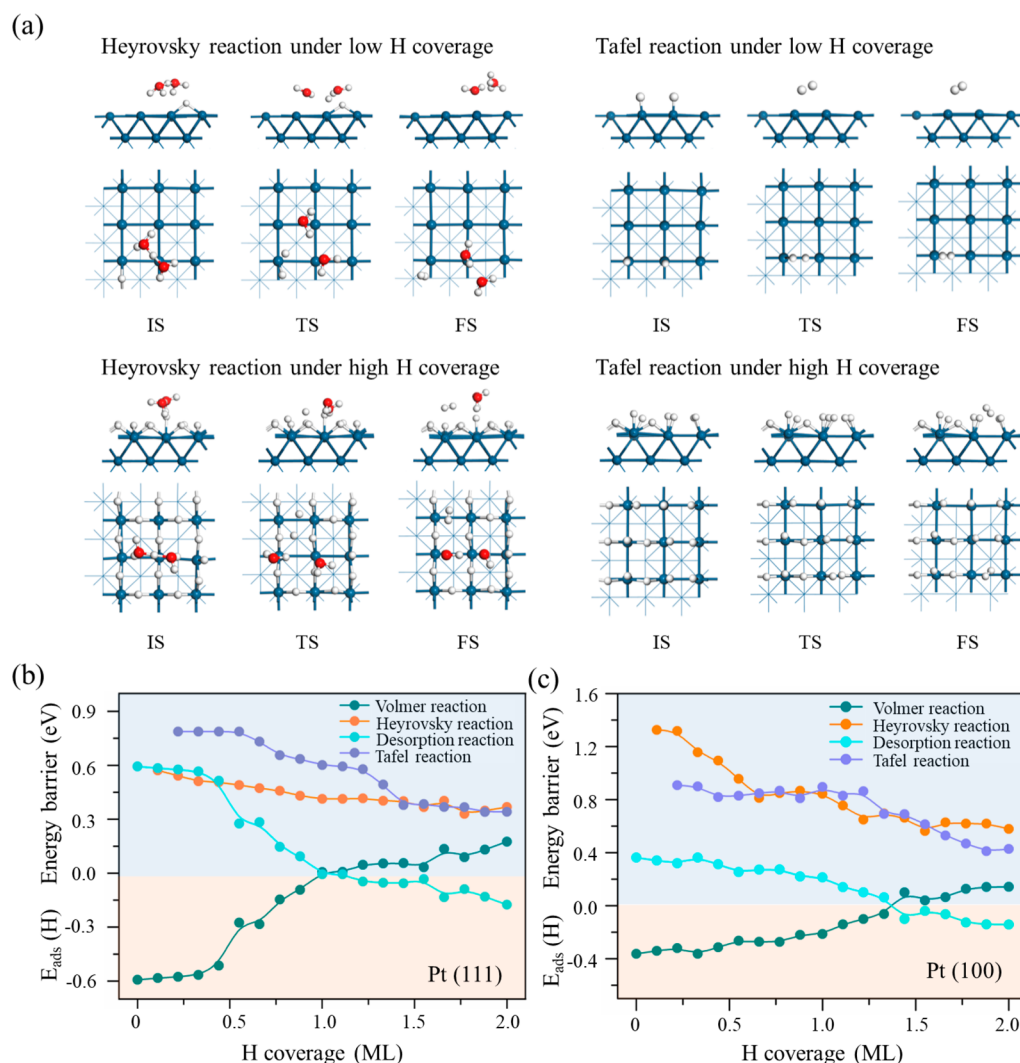


Figure 4. (a) Typical optimized configurations for the reaction transition states of Heyrovsky and Tafel reactions under low and high hydrogen coverage. IS, FS and TS represent the initial state, final state and the transition state, respectively. DFT-calculated energy barriers and adsorption/reaction energies of HER elementary reactions on (b) Pt (111) and (c) Pt (100) surfaces at different H coverage.

$$\Delta t = -\frac{R}{\ln \rho_2} \quad (9)$$

where R is the total rate, ρ_2 is the random number, both are generated in step (iii). Subsequently, the total time updated by incorporating the time interval Δt and the current time t as follows:

$$t = t + \Delta t$$

- (vii) Checking whether the simulation reach the stop conditions, and terminate the simulation if $t > t_{\text{max}}$; otherwise, turn back to step (ii).

In this study, the encoded KMC simulation was conducted using MATLAB. In order to ensure convergence and accuracy of the results, all of the simulations were performed for more than 10^8 KMC steps; the time step was determined by rate constant and random number ρ_2 (eq 9). The simulation potential window was set from 0.05 to -0.40 V with a 0.05 interval. Selecting an appropriate lattice size is crucial for KMC simulations. Large lattice offers more accurate representation of the catalyst surface, but consumes more CPU time. Therefore, it is important to find a balance between lattice

size and computational efficiency. In this work, the effect of lattice sizes of $N = 50, 100, 250$, and 500 is assessed, and the results are presented in Figure S2. The results suggest that a lattice size of 50 yields a relatively large deviation, whereas calculations under lattice sizes of $100, 250$, and 500 exhibited similar results. Therefore, a lattice size of $N = 100$ is sufficient to ensure the model accuracy with acceptable computational resource.

The current density (j) is correlated to the number of the produced H_2 molecule, $n(\text{H}_2, \text{mol})$, which can be tracked by using the proposed KMC model, i.e.,

$$n_e = 2n(\text{H}_2) \quad (10)$$

$$I = \frac{Q}{t} = \frac{F \times n_e}{t} = \frac{2F \times n(\text{H}_2)}{t} \quad (11)$$

$$j = \frac{I}{A} = \frac{2F \times n(\text{H}_2)}{t \times A} \quad (12)$$

where t is the KMC simulation time (s), n_e is the number of transferred electrons in HER (mol), I is the current (A), Q is the total quantity of electric charge (C). A represents the

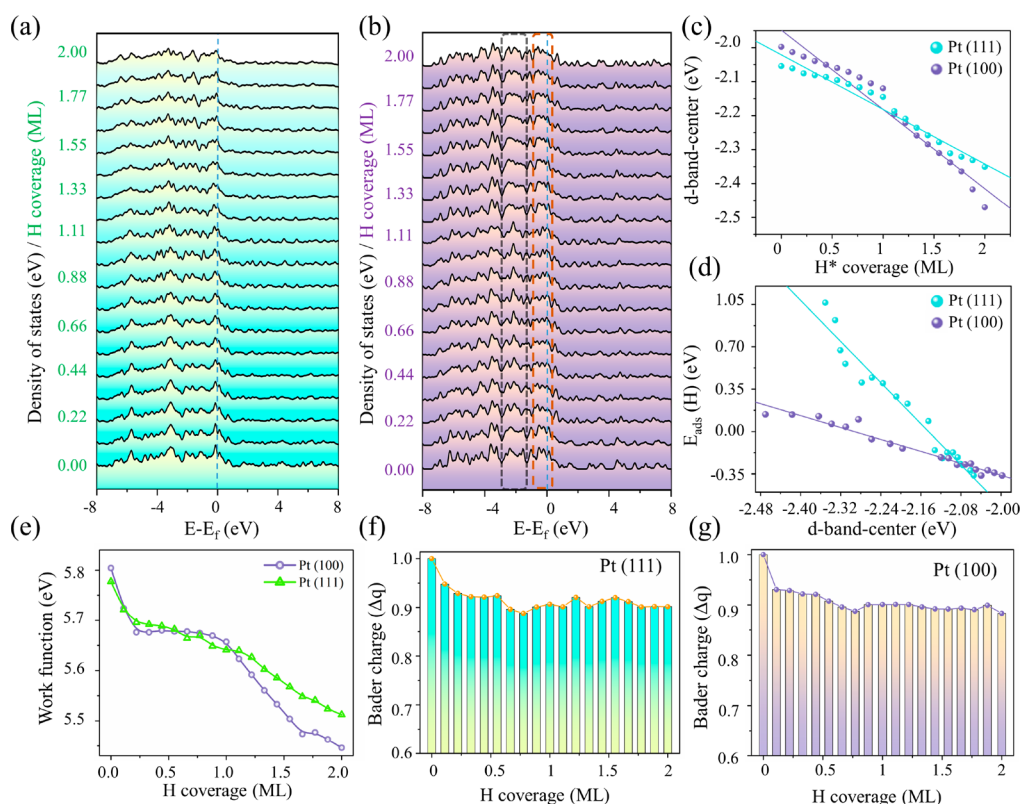


Figure 5. (a) Projected density of states (PDOS) on Pt 5d orbital at Pt (111) and (b) Pt (100) at different H coverage, (c) d-band center for Pt (111) and Pt (100), (d) linear plot of d-band-center and $E_{\text{ads}}(\text{H})$, (e) work functions of Pt (111) and Pt (100) surface under different H coverage, and (f) surface H atom charge on Pt (111) and (g) Pt (100).

surface area (cm^{-2}), and F is the Faraday constant (96485 C mol^{-1}). By integrating the current density occurring during the KMC simulation time intervals at different working potentials, the HER LSV curve can be obtained.

2.5. Details for Electrochemical and Spectroscopic Measurements. To validate the proposed DFT-KMC model, electrochemical experiments were performed in a standard three-electrode system on a CHI760E electrochemical workstation. The glassy carbon electrode coated with commercial 20 wt % Pt/C catalyst was employed as working electrode; the electrolyte is 0.1 M HClO_4 , whereas the scan rate of LSV is set at 5 mVs^{-1} . The Ag/AgCl electrode and carbon rod was used as reference and counter electrode, respectively. The measured potentials were calibrated to the reversible hydrogen electrode (RHE) by the equation: $E_{\text{RHE}} = E_{\text{Ag/AgCl}} + 0.197 + 0.059 \times \text{pH}$. The *in situ* attenuated total reflection Fourier transformed infrared spectroscopy (ATR-FTIR) was carried out on a INVENIO S spectrometer equipped with an MCT detector. The Si cylindrical prism (20 mm in diameter) was used as the conductive substrate for catalysts and the IR reflection element. The Pt catalyst suspension was dropped on the Si surface as the working electrode. The loading of the catalyst was 1 mg cm^{-2} , and the electrolyte was 0.1 M HClO_4 .

3. RESULTS AND DISCUSSION

3.1. DFT Calculation Results. 3.1.1. HER Kinetic Calculations. The energy barriers of HER elementary reactions on Pt (111) and Pt (100) under different H coverages were obtained from DFT calculations. Figure 4a illustrates the typical configurations of the reaction transition states of Heyrovsky and Tafel reactions under low and high H

coverage. Figure 4b-c represents the different reaction energy barriers at increasing hydrogen coverage. All the optimized configurations of the HER reaction transition states under different H coverages are presented in Figures S3–S12. Noted that, in this study, the energy barrier calculations are performed on the basis of the optimized surface configurations under various catalyst surface adsorption and electrolyte presence states. There are many possible spatial surface configurations under a specific surface hydrogen coverage value. However, the calculations show that under the same H converge, the H spatial configurations have a very minor impact on the energetics of surface H atoms (Table S1). Also, a previous study reported that the H diffusion barrier on Pt is almost negligible.^{46,47} Accordingly, it is expected that, under a specific hydrogen coverage, the effect of spatial distribution of surface H atoms on their pairwise interactions and the HER energetics should be minor. Therefore, for a specific hydrogen coverage value, the most stable catalyst surface coverage state is identified first, followed by the calculation of the HER energies. Although the microenvironment of a catalyst surface state or the H spatial configurations would be very complex, the adoption of this relaxed catalyst surface state is still reasonable since the relaxed catalyst surface state would also be the most likely occurrence state in the course of long-term dynamic evolution of HER. The energy barrier of the Volmer reaction is taken as the adsorption energy. When the reaction energy is negative, the energy barrier for the Volmer reaction is set to 0. This is because the Volmer reaction is an adsorption process, and negative reaction energy values suggest that it would readily occur during HER.^{48,49} Therefore, in Figure 4b-c, the region below 0 represents the reaction energy of the

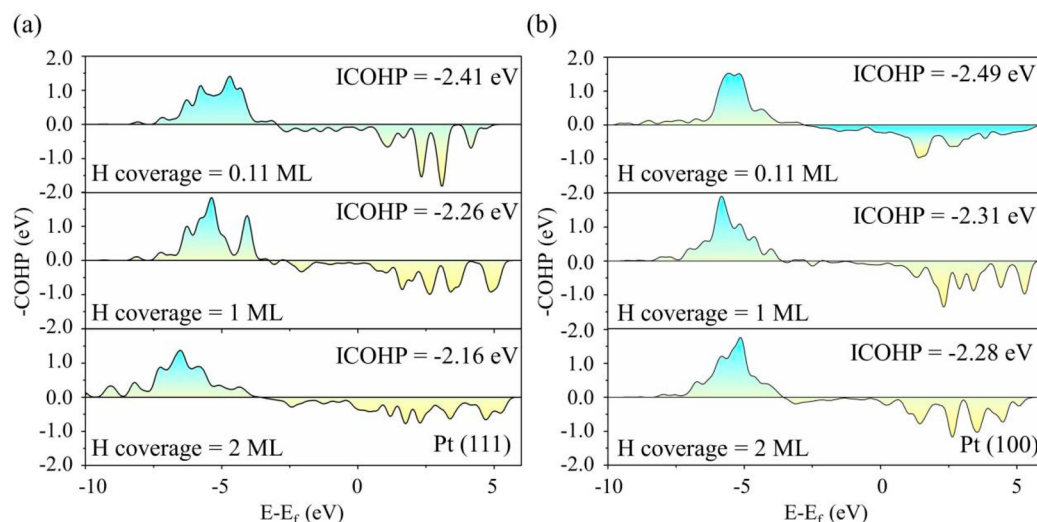


Figure 6. -COHP and ICOHP between the surface H and Pt atom on different H coverages on Pt(111) (a) and Pt (100) (b).

Volmer reaction, whereas the positive region represents the energy barrier of the reaction steps. One can observe that, on the Pt(111) surface, the Heyrovsky reaction barrier is lower than that of the Tafel reaction under the 1.25 H* ML range (Figure 4b). A further increase of H coverage would lead to slightly higher activation barrier for the Heyrovsky reaction than Tafel reaction. In addition, the energy barrier for Tafel reaction on Pt (111) is in the range of 0.45–0.80 eV, which is also in agreement with previous experimental results.⁵⁰ Additionally, the Volmer reaction on Pt (111) surface changes from exothermic to endothermic at around 1 H* ML. Meanwhile, the Volmer reaction step becomes endothermic by 0.01 eV. This implies that the H coverage on Pt (111) is about 1 H* ML at the equilibrium potential, which is consistent with a previous experimental study.⁵¹ Similarly, on the Pt (100) surface, the Volmer reaction is exothermic until the H coverage reaches approximately 1.4 H* ML (Figure 4c). This suggests that the total coverage of H atom on Pt (100) is about 1.4 ML at the equilibrium potential. This result is consistent with that (1.3 ML) reported by Strmcnik et al.⁵² After that, the reaction barrier gradually increases to 0.24 eV as the H coverage increases up to 2 H* ML. Note that the energy barrier of the Volmer reaction increases with the hydrogen coverage on both Pt (111) and Pt (100). This can be attributed to the growing H–H repulsive interaction at higher H coverage. The energy barrier of the Heyrovsky and Tafel reaction on both Pt (111) and Pt (100) exhibits a decreasing trend with the increased H coverage. This can be attributed to the weakening of the Pt–H bond, which is discussed in Section 3.1.2. On Pt (100), Heyrovsky reaction is more sensitive toward H coverage, showing a faster decreasing rate than Tafel reaction. Nevertheless, even under 2H* ML, the Tafel reaction still exhibits lower energy barriers than the Heyrovsky reaction. These results suggest that the sensitivity of the reaction energy barrier toward H coverage varies with Pt surfaces. On the Pt(111) surface, the impact of H coverage is most significant for the energetics of Volmer reaction. On Pt (100), the most sensitive step is the Heyrovsky reaction. For the Volmer reaction, a significant change from exothermic to endothermic is observed even as the H coverage increases. These findings illustrate the significance of considering the influence of H coverage (0–2 ML) in the mechanistic study of the HER

process, which has been usually neglected in the open literature.

3.1.2. Electronic Structure Analysis. In order to provide an in-depth understanding of HER energetics, the electronic structure of the Pt (111) and Pt (100) surfaces under different H coverages is investigated. The projected density of states (PDOS) of Pt d orbital in 0 to 2 H* ML H-coverage range is shown in Figure 5. As shown in this figure, as the coverage increases on Pt (111), the primary Pt 5d sharp peak near the Fermi level shifts toward the valence band (Figure 5a). Simultaneously, the adjacent small sharp peak gradually disappears. These observations suggest that, on the Pt (111) surface, the electronic states of Pt d orbital gradually approach the occupied states as the H coverage increases, and the d-band center shifts downward. Similar conclusions can be drawn for the Pt (100) surface (Figure 5b). Notably, when the H coverage exceeds 1 ML, the sharp Pt 5d peak (black dashed border) located at the left of the main peak near the Fermi level (red dashed border) becomes smooth. This suggests that, with the increase of H coverage, the electrons on Pt (100) surface tend to become more delocalized.⁵³ That is, the Pt (100) surface with high H coverage is more prone to an electronic redistribution, which is beneficial for HER performance.⁵⁴ Figure 5c shows that the Pt(111) and Pt (100) d-band-center rapidly decreases with the increase of H coverage. This can be explained by the hybridization between the H 1s and Pt 5d orbitals. That is, upon the H adsorption on clean Pt sites, the hybridization of the Pt 5d and H 1s orbitals occurs and promotes partial electron filling of the d-states. Meanwhile, the apparent increased filling of the d-states would result into rapid decrease of the d-band center, weakening the Pt–H bonding.⁵⁵ Figure 5d shows that there is an approximated linear relationship between the H adsorption energy and d-band-center, showing that H_{ads} gradually decrease with the d-band-center.

Figure 5e presents the work functions of Pt (111) and Pt (100) under different H surface coverages. As shown in this figure, the work function of Pt catalyst gradually decreases as the H coverage increases. Similar trends have been observed in experimental measurements.⁵⁶ The lower work function enables the catalyst with a rapid electron transfer property, which would promote the HER process.⁵³ It can be seen that the Pt

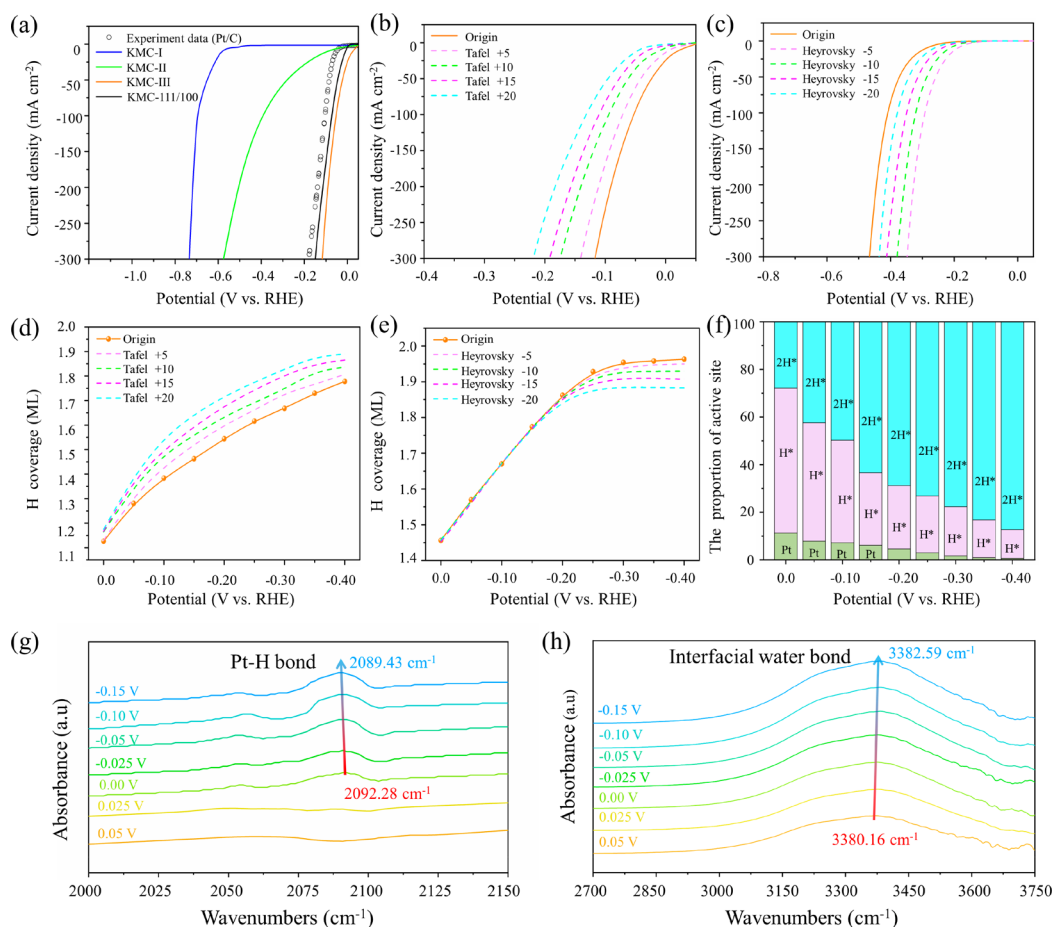


Figure 7. (a) Comparison of KMC predicted HER LSV curves on Pt (111) under different conditions (KMC-I, KMC-II, KMC-III and KMC-111/100) with experimental data. (b) Sensitivity analysis for the effect of Tafel reaction energetics on HER LSV for Pt (111), and (c) the effect of Heyrovsky reaction energetics on HER LSV for Pt (100). (d) Sensitivity analysis of Tafel reaction energetic effect on the H coverage for Pt (111) and (e) effect of Heyrovsky reaction energetic on the H coverage for Pt (100). (f) The evolution of surface species on Pt (111) at different HER working potentials. (g) *In situ* ATR-FTIR spectra recorded from 0.05 to -0.15 V vs RHE in 0.1 M HClO_4 for Pt–H bond and (h) interfacial water molecules analysis.

(100) surface starts with a work function higher than that of Pt (111), and as the H coverage increases, the work functions of both models decreased significantly. The work function of Pt (100) surface shows a higher sensitivity to the H coverage than Pt (111), and reaches a lower work function at high H coverage. The change of work function further confirms that there exist an electron redistribution due to the increased H coverage.⁵⁷ Figure 5f,g shows the average charge of the H_{ads} exhibiting an overall decreasing trend with increasing H coverage on both Pt (111) and Pt (100) surfaces. Notably, the electron transfer on Pt (100) is more pronounced, which is consistent with the fast decrease in its work function.

To further evaluate the strength of Pt–H bonding under different H surface coverages, the crystal orbital Hamilton population (COHP) of Pt–H bonding is depicted in Figure 6. The upward peaks (positive y-values) represent bonding interactions, whereas the downward peaks (negative y-values) indicate antibonding interactions. The integrated crystal orbital Hamilton population (ICOHP) is also calculated for quantitative comparison.⁵⁸ A more negative ICOHP value indicates a stronger binding strength. Figure 6a shows that the ICOHP for Pt–H bonds on Pt (111) at 0.11 H^* ML is -2.41 eV, which is higher than that at 1 H^* ML (-2.26 eV) and 2 H^* ML coverage (-2.16 eV). The ICOHP values indicate an

increase in the antibonding contributions at higher H coverage. This provides evidence for the weakening of Pt–H bonds with increased H surface coverage. Similar conclusions were drawn for the Pt (100) surface (Figure 6b), where the ICOHP value at 0.11 H^* ML (-2.49 eV) is higher than that at 1 H^* ML (-2.31 eV) and 2 H^* ML (-2.28 eV). Overall, the electronic structure calculation provides theoretical explanations on the HER energetic results under different H-coverage.

3.2. KMC Simulation Results. The LSV curve for the HER reaction is predicted by KMC simulation, and the comparison with that of Pt/C catalysts can serve as a good validation of the proposed KMC model (Figure 7a). The KMC simulated LSV curves considering 1 H^* ML coverage and absence of H_5O_2^+ ion are also presented. These include the LSV curves that *i*) only considered the H^* ML coverage changes in the range of 0 – 1 (KMC-I); *ii*) H coverage change in the 0 – 2H^* ML range, but the Volmer reaction was calculated by H^+ rather than H_5O_2^+ (KMC-II); *iii*) H coverage change in the range of 0 – 2H^* ML and the Volmer reaction was calculated in the presence of H_5O_2^+ (KMC-III). The events list used for KMC-I to KMC-III models was presented in Table S2, and their kinetic parameters were shown in Tables S3–S5. In Figure 7a, it can be seen that there is a significant inconsistency between the KMC-I predictions and experiment

data. This suggests that H coverage and solvent effect should be considered in the modeling of HER process. Considering high H coverage condition would predict an improved KMC-II LSV curve. However, due to the absence of considering the H_2O_2^+ presence in the Volmer reaction, the predictions still show a considerable deviation from the experimental results. The KMC-III LSV curve is much closer to the experimental results, demonstrating that the consideration of high H coverage range and H_2O_2^+ presence in Volmer reaction is crucial for realistic modeling of the HER reaction. In addition, note that the Pt (111) and Pt (100) crystal planes coexist in Pt/C catalysts. It has been reported that the largest surface area in Pt nanocatalyst has a (100)/(111) intensity ratio of approximately 40.4%.⁵⁹ Hence, the LSV curve contributed by Pt (111) and Pt (100) at this proportion is also presented in Figure 7a (KMC-111/100). As shown in this figure, this KMC-111/100 LSV curve agrees with the experimental data, further confirming the accuracy of the proposed KMC model.

The sensitivity of the DFT energetics of the HER steps to the simulated LSV curves was examined. For instance, “+10” and “-10” represent a 10% increase and decrease in the energy barrier of a specific HER step, respectively. One can observe that, on Pt (111), the LSV curve is very sensitive to the Tafel reaction barrier, indicating that HER activity is mainly driven by Tafel reaction (Figure 7b). Similar sensitivity analyses for the Heyrovsky and Volmer reaction barriers are presented in Figure S13. It is found that the LSV curves are more sensitive for the energy barriers of the Tafel reaction rather than the Heyrovsky and Volmer reactions. This suggests that the Tafel step might be the controlling step for the HER on Pt (111). This agrees with the study of Marković et al.⁶⁰ and Gómez et al.,⁶¹ which reported that HER follows Volmer–Tafel pathways on Pt (111). However, the study of Nørskov et al.⁵¹ found that Heyrovsky reaction might be the dominating process on Pt (111) due to its relatively lower energy barrier. However, in KMC simulation, the occurrence of HER steps not only depends on the intrinsic energetics but also is governed by the local coordination environment of the reactants. For example, in KMC, the Heyrovsky reaction occurs between the H_2O_2^+ in the solvent and the surface Pt-H* or Pt-2H*. The Tafel reaction occurs by the formation of adjacent Pt-H* or Pt-2H* sites. Accordingly, the Tafel reaction, as a surface reaction process, can be greatly affected by the local surface environment compared to the Heyrovsky reaction. Therefore, at high H coverage, the increased surface H* sites might promote the occurrence of the Tafel reaction. This suggests that the Tafel reaction might become the dominant process under high H coverage conditions, despite its relatively high energy barrier when compared to the Heyrovsky reaction. Figure 7c presents the KMC predicted HER LSV curve on Pt (100) surface and its sensitivity toward the Heyrovsky reaction energy barrier. The sensitivity analysis of the Tafel and Volmer reaction barriers is shown in Figure S14. The results show that, on the Pt (100) surface, the overall HER activity is lower than Pt (111), and its performance is very sensitive to the energy barrier of Heyrovsky reaction followed by Tafel and Volmer reactions. This indicates that kinetics of HER reaction on Pt (100) surface is mainly dominated by the Heyrovsky reaction.

Figure 7d illustrates the sensitivity of the H surface coverage under different Tafel reaction barriers and working potentials. This figure shows that the increase in applied potential leads to a significant increase of the H surface coverage, highlighting

the importance of considering dynamic H coverage in HER mechanistic studies. In addition, with the increase of Tafel reaction barrier, the H coverage on the catalyst surface decreases. This behavior is expected since the high energy barrier of Tafel reaction would slow down the consumption of surface H*. Moreover, a sensitivity analysis of H coverage toward Heyrovsky reaction barrier at different applied voltages was also examined (Figure 7e). The results from this analysis showed that Pt(100) surface exhibits a higher surface H coverage at the same working potential compared to Pt(111). This can be attributed to the lower Volmer reaction barrier on Pt(100) than Pt(111). In contrast to that observed on Pt (111), the H coverage on Pt (100) tends to reach an equilibrium state as the applied potential increases from −0.30 V to −0.40 V vs RHE. This might be attributed to the higher H coverage causing the increase in the Volmer reaction barrier. This inhibits the promoting effect of increasing working potential on the Volmer reaction. Therefore, Pt (100) surfaces exhibit a higher increase rate in H coverage at lower working potential, whereas the increase of H coverage slows down at higher working potential. In addition, it can also be observed that the equilibrium state of Pt (100) occurs at high H coverage around 1.9 ML, where most of the surface sites are occupied by [Pt-2H] species. At this state, the reaction rate of the Volmer reaction significantly decreases, and the promoting effect of higher working potential is suppressed. Note that, due to the high Volmer reaction barrier on Pt (111) surface, the H coverage only reaches 1.75 ML at the operation potential −0.30 V to −0.40 V vs RHE. At this state, the Volmer reaction can be further promoted by increasing the HER working potential. This provides an explanation for the absence of H coverage equilibrium on the Pt (111) surface in −0.30 V to −0.40 V vs RHE range. Figure 7f shows the distribution of surface species at different applied voltages obtained using a 10% increase in the Tafel reaction barrier on Pt (111) surface. This energy barrier was chosen because the KMC simulation results fits very well with the experimental results. One can see that, with the increasing working potential, the [Pt-2H] species increases, lowering the number of [Pt-H] species and vacant Pt sites. This observation suggests that, as the working potential increases, the H coverage on the catalyst surface also increases. Consequently, the energy barrier for the protonation reaction decreases, leading to enhanced HER activity. This has also been observed in a previous experimental study,⁶² which reported that the increase in working potential leads to decrease in the dipole moment of the Pt–H surface and the Pt–H stretching frequency. This would promote the Volmer reaction, resulting in an increase of H coverage.¹⁹ In another theoretical study, Abghoui et al.⁶³ found that the Volmer reaction barrier tends to decrease with higher working potential. That is, a higher working potential promotes HER by lowering the Volmer reaction barrier, which results in an increase in H coverage. The KMC predictions agree with those reported in previous studies and experimentally measured HER LSV data. This suggests the as-proposed KMC model captures the catalytic HER reaction on the Pt catalyst.

To provide a direct support on the KMC predicted HER evolution behavior on Pt catalyst, *in situ* ATR-FTIR analysis was conducted to monitor the H binding strength on Pt and the interfacial water molecules binding state. As shown in Figure 7g, the peak centered at 2092.28 cm^{-1} is attributed by the Pt–H bond.^{62,64} It can be seen that the peaks of Pt–H bond show a significant red shift along with the increased

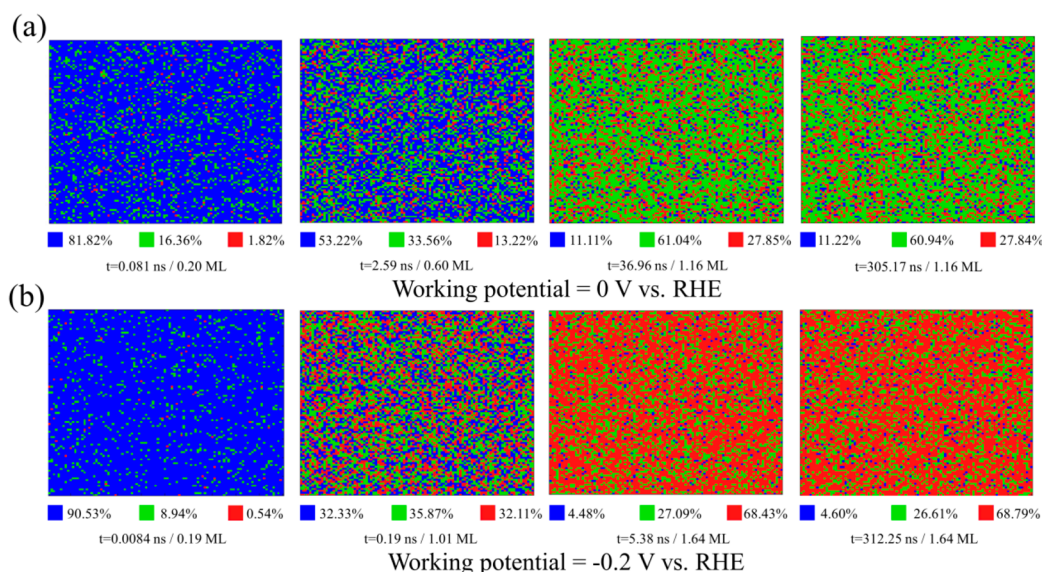


Figure 8. Snapshot of the steady-state Pt (111) surface HER species under different working potentials: 0 V (a) and -0.2 V vs RHE (b). Blue points represent the [Pt] sites, the green points represent the [Pt-H] sites, and the red points represent the [Pt-2H] sites.

working potentials and reached 2089.43 cm^{-1} at -0.15 V vs RHE. The decreased wavenumber indicates a weakened Pt–H bonding with the increasing working potential. Moreover, the peak within $3380.16\text{--}3382.59\text{ cm}^{-1}$ in Figure 7h is attributed to the hydrogen-bonded interfacial water molecules.⁶⁵ One can see that the wavenumbers of the interfacial water peak generally increase with the increasing working potential. The increased vibrational mode wavenumber of the interfacial water molecules implies a weakened interaction between Pt and the interface water molecules.⁶⁴ Accordingly, it is expected that more surface active sites would be occupied by H^* rather than water molecules at higher HER working potentials, leading to increased H coverage. These observations are consistent with the DFT calculations and KMC simulation results. That is, the high working potentials would lead to increased H surface coverage. This would lower the d-band center of Pt and weaken the Pt–H binding energy, thus promoting HER kinetics.

Figure 8 presents the KMC predicted HER surface species evolution on the Pt(111) surface at different reaction times and working potentials. Figure 8a depicts the surface species evolution process at 0 V vs RHE. As shown in this figure, at the initial stage of the HER process (0.081 ns), the Pt (111) surface mainly consists of vacant Pt sites (81.82%), followed by [Pt-H] sites (16.36%) and [Pt-2H] site (1.82%). The total H coverage is 0.20 ML. As the reaction continues, the vacant [Pt] sites decreases to 53.22% at 2.59 ns, and the [Pt-H] and [Pt-2H] sites increased to 33.56% and 13.22%, respectively, corresponding to a H total coverage of 0.60 ML. At 36.96 ns, the surface reaches a dynamic equilibrium state, showing similar H coverage with that at 305.17 ns. This suggests that the overall H coverage tends to stabilize at 1.16 ML. Similarly, the evolution of surface active species at -0.2 V vs RHE working potential is shown on Figure 8b. Under -0.2 V vs RHE operation condition, it takes only about 0.19 ns for the H surface coverage to reach 1.01 ML. After that, the Pt (111) surface reached a dynamic equilibrium state at around 5.38 ns, much faster than that under 0 V vs RHE (36.96 ns). Moreover, the equilibrium H coverage (1.64 ML) at -0.2 V vs RHE is much higher than 0 V vs RHE (1.16 ML). These results show

that, under higher working voltages, the catalyst surface tends to reach a dynamic equilibrium between the production and consumption of H^* in the HER process more rapidly with higher H coverage.

4. CONCLUSION

In this study, a first-principles-based coverage and potential-dependent KMC model of HER reaction on Pt (111) and Pt (100) surfaces was presented. The HER kinetics were first investigated by DFT calculations considering the presence of H_3O_2^+ solvent, H surface coverage and electric field effects. Electronic structure analysis demonstrates that the higher H surface can lead to a downward shift of the d-band center of Pt (111) and Pt (100) and the Pt–H bond is weakened, leading to a decrease in the protonation reaction barrier with promoted HER kinetics. Based on the DFT results, a KMC model was developed to simulate the evolution of the HER reaction on Pt catalyst surfaces. Accurate predictions of the HER polarization curve is achieved using the KMC model, demonstrating the consideration of H_3O_2^+ species, and dynamic evaluation of H surface coverage is essential for HER mechanistic study. Sensitivity analysis of the KMC predicted polarization curve reveals that HER on Pt(111) surface is mainly governed by the Tafel pathway, whereas HER on the Pt(100) surface is primarily governed by the Heyrovsky pathway. Moreover, the analysis of surface species evolution demonstrates that, with increasing working potential, there is an accelerated formation of [Pt-2H] species, leading to an increase in H coverage and consequently promoting the HER process. The predicted weakened Pt–H binding strength and increased H coverage at high HER working potential was also supported by *in situ* ATR-FTIR spectroscopy analysis. Therefore, the proposed DFT-KMC model provides a unique dynamic perspective for comprehensive understanding of the electrochemical HER mechanism under realistic operating conditions.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.3c04588>.

Additional details of convergence test, DFT calculation configuration, kinetic parameters and KMC simulation results (PDF)

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Notes

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■ REFERENCES

- (1) Finegold, L.; Cude, J. L. Biological Sciences: One and Two-Dimensional Structure of Alpha-Helix and Beta-Sheet Forms of Poly(L-Alanine) Shown by Specific Heat Measurements at Low Temperatures (1.5–20 K). *Nature* **1972**, 238 (5358), 38–40.
- (2) Ren, B.; Jin, Q.; Li, Y.; Li, Y.; Cui, H.; Wang, C. Activating Titanium Dioxide as a New Efficient Electrocatalyst: From Theory to Experiment. *ACS Appl. Mater. Interfaces* **2020**, 12 (10), 11607–11615.
- (3) Yu, P.; Wang, F.; Shifa, T. A.; Zhan, X.; Lou, X.; Xia, F.; He, J. Earth Abundant Materials beyond Transition Metal Dichalcogenides: A Focus on Electrocatalyzing Hydrogen Evolution Reaction. *Nano Energy* **2019**, 58 (December 2018), 244–276.
- (4) Di, J.; Chen, C.; Zhu, C.; Long, R.; Chen, H.; Cao, X.; Xiong, J.; Weng, Y.; Song, L.; Li, S.; Li, H.; Xiong, Y.; Liu, Z. Surface Local Polarization Induced by Bismuth-Oxygen Vacancy Pairs Tuning Non-Covalent Interaction for CO₂ Photoreduction. *Adv. Energy Mater.* **2021**, 11, 2102389.
- (5) Bao, F.; Kempainen, E.; Dorbandt, I.; Bors, R.; Xi, F.; Schlattmann, R.; van de Krol, R.; Calnan, S. Understanding the Hydrogen Evolution Reaction Kinetics of Electrodeposited Nickel-Molybdenum in Acidic, Near-Neutral, and Alkaline Conditions. *ChemElectroChem* **2021**, 8 (1), 195–208.
- (6) Yu, G. Q.; Huang, B. Y.; Chen, S. M.; Liao, J. W.; Yin, W. J.; Teobaldi, G.; Li, X. B. The Combined Role of Faceting and Heteroatom Doping for Hydrogen Evolution on a WC Electrocatalyst in Aqueous Solution: A Density Functional Theory Study. *J. Phys. Chem. C* **2021**, 125 (8), 4602–4613.
- (7) Ekspong, J.; Gracia-Espino, E.; Wagberg, T. Hydrogen Evolution Reaction Activity of Heterogeneous Materials: A Theoretical Model. *J. Phys. Chem. C* **2020**, 124 (38), 20911–20921.
- (8) Qin, X.; Zhu, S.; Wang, Y.; Pan, D.; Shao, M. Full Atomistic Mechanism Study of Hydrogen Evolution Reaction on Pt Surfaces at Universal PHs: Ab Initio Simulations at Electrochemical Interfaces. *Electrochim. Acta* **2022**, 425 (June), No. 140709.
- (9) Shi, Q.; Sun, R. Adsorption Manners of Hydrogen on Pt(1 0 0), (1 1 0) and (1 1 1) Surfaces at High Coverage. *Comput. Theor. Chem.* **2017**, 1106, 43–49.
- (10) Hamada, I.; Morikawa, Y. Density-Functional Analysis of Hydrogen on Pt(111): Electric Field, Solvent, and Coverage Effects. *J. Phys. Chem. C* **2008**, 112 (44), 17486.
- (11) Exner, K. S.; Over, H. Kinetics of Electrocatalytic Reactions from First-Principles: A Critical Comparison with the Ab Initio Thermodynamics Approach. *Acc. Chem. Res.* **2017**, 50 (5), 1240–1247.
- (12) Lee, C. H.; Pahari, S.; Sitapure, N.; Barteau, M. A.; Kwon, J. S. Investigating High-Performance Non-Precious Transition Metal Oxide Catalysts for Nitrogen Reduction Reaction: A Multifaceted DFT – KMC – LSTM Approach. *ACS Catal.* **2023**, 13, 8336.
- (13) Lee, C. H.; Pahari, S.; Sitapure, N.; Barteau, M. A.; Kwon, J. S. II. DFT-KMC Analysis for Identifying Novel Bimetallic Electrocatalysts for Enhanced NRR Performance by Suppressing HER at Ambient Conditions Via Active-Site Separation. *ACS Catal.* **2022**, 12 (24), 15609–15617.
- (14) Yuan, Y.; Dong, X.; Ricardez-Sandoval, L. A Multi-Scale Model for Syngas Combustion on NiO Oxygen Carrier for Chemical Looping Combustion: The Role of Nearest Neighbors. *Fuel Process. Technol.* **2022**, 229 (December 2021), No. 107172.
- (15) Ricardez-Sandoval, L. A. Current Challenges in the Design and Control of Multiscale Systems. *Can. J. Chem. Eng.* **2011**, 89 (6), 1324–1341.
- (16) Kim, K. Y.; Lee, J.; Kang, S.; Son, Y. W.; Jang, H. W.; Kang, Y.; Han, S. Role of Hyper-Reduced States in Hydrogen Evolution Reaction at Sulfur Vacancy in MoS₂. *ACS Catal.* **2018**, 8 (5), 4508–4515.
- (17) Vener, M. V.; Librovich, N. B. The Structure and Vibrational Spectra of Proton Hydrates: H₃O₂⁺ as a Simplest Stable Ion. *Int. Rev. Phys. Chem.* **2009**, 28 (3), 407–434.
- (18) Pašti, I. A.; Leetmaa, M.; Skorodumova, N. V. General Principles for Designing Supported Catalysts for Hydrogen Evolution Reaction Based on Conceptual Kinetic Monte Carlo Modeling. *Int. J. Hydrogen Energy* **2016**, 41 (4), 2526–2538.

- (19) Kronberg, R.; Laasonen, K. Reconciling the Experimental and Computational Hydrogen Evolution Activities of Pt(111) through DFT-Based Constrained MD Simulations. *ACS Catal.* **2021**, *11* (11), 8062–8078.
- (20) Zhao, L.; Liu, H.; Liu, Y.; Han, X.; Xu, J.; Xing, W.; Guo, W. Mechanistic Insights into the Hydrogen Oxidation Reaction on PtNi Alloys in Alkaline Media: A First-Principles Investigation. *ACS Appl. Mater. Interfaces* **2020**, *12* (36), 40248–40260.
- (21) Petrova, N. V.; Yakovkin, I. N. Hydrogen Associative Desorption from Ru(1010). *Eur. Phys. J. B* **2008**, *63* (1), 17–24.
- (22) Gossenberger, F.; Juarez, F.; Groß, A. Sulfate, Bisulfate, and Hydrogen Co-Adsorption on Pt(111) and Au(111) in an Electrochemical Environment. *Front. Chem.* **2020**, *8* (July), 1–11.
- (23) Santana, J. A. DFT Calculations of the Adsorption States of O₂ on OH/H₂O-Covered Pt(111). *Electrocatalysis* **2020**, *11* (6), 612–617.
- (24) Li, K.; Li, Y.; Wang, Y.; He, F.; Jiao, M.; Tang, H.; Wu, Z. Oxygen Reduction Reaction on Pt(111) and Pt(100) Surfaces Substituted by the Subsurface Cu: A Theoretical Perspective. *J. Mater. Chem. A* **2015**, *3*, 11444.
- (25) Duan, Z.; Wang, G. Comparison of Reaction Energetics for Oxygen Reduction Reactions on Pt(100), Pt(111), Pt/Ni(100), and Pt/Ni(111) Surfaces: A First-Principles Study. *J. Phys. Chem. C* **2013**, *117* (12), 6284–6292.
- (26) Chou, H. L.; Rick, J. Investigation of CO and OH Adsorption and Oxidation in the Presence of Cocatalytic Ruthenium Ions on the Pt(111) Surface. *Catal. Commun.* **2022**, *162*, No. 106400.
- (27) Li, L.; Chen, H.; Li, L.; Li, B.; Wu, Q.; Cui, C.; Deng, B.; Luo, Y.; Liu, Q.; Li, T.; Zhang, F.; Asiri, A. M.; Feng, Z. S.; Wang, Y.; Sun, X. La-Doped TiO₂ Nanorods toward Boosted Electrocatalytic N₂-to-NH₃ Conversion at Ambient Conditions. *Chin. J. Catal.* **2021**, *42* (10), 1755–1762.
- (28) Chen, H.; Zheng, T.; He, Q.; Shang, L.; Wang, G.; Wang, Q.; Wang, X.; Shi, X.; Gu, M.; Jiang, Z. Local Coordination and Ordering Engineering to Design Efficient Core-Shell Oxygen Reduction Catalysts. *J. Electrochem. Soc.* **2020**, *167* (14), No. 144501.
- (29) Tan, T. L.; Wang, L. L.; Johnson, D. D.; Bai, K. Hydrogen Deposition on Pt(111) during Electrochemical Hydrogen Evolution from a First-Principles Multiadsorption-Site Study. *J. Phys. Chem. C* **2013**, *117* (44), 22696–22704.
- (30) Yan, L.; Sun, Y.; Yamamoto, Y.; Kasamatsu, S.; Hamada, I.; Sugino, O. Hydrogen Adsorption on Pt(111) Revisited from Random Phase Approximation. *J. Chem. Phys.* **2018**, *149* (16), 164702.
- (31) Du, X.; Liu, G.; Luo, Y.; Li, J.; Ricardez-Sandoval, L. Theoretical Insights into the Oxygen Reduction Reaction on PtCu (1 1 1): Effects of Surface Defect and Acidic Solvent. *Appl. Surf. Sci.* **2021**, *570* (August), No. 151195.
- (32) Vasylenko, A.; Gamon, J.; Duff, B. B.; Gusev, V. V.; Daniels, L. M.; Zanella, M.; Shin, J. F.; Sharp, P. M.; Morscher, A.; Chen, R.; Neale, A. R.; Hardwick, L. J.; Claridge, J. B.; Blanc, F.; Gaultois, M. W.; Dyer, M. S.; Rosseinsky, M. J. Element Selection for Crystalline Inorganic Solid Discovery Guided by Unsupervised Machine Learning of Experimentally Explored Chemistry. *Nat. Commun.* **2021**, *12*, 1–12.
- (33) Kresse, G.; Furthmüller, J. Efficiency of Ab-Initio Total Energy Calculations for Metals and Semiconductors Using a Plane-Wave Basis Set. *Comput. Mater. Sci.* **1996**, *6* (1), 15–50.
- (34) Sadeghian, S.; Jafari, M. Density Functional Investigation of Oxygen Reduction Reaction on Pt₃Pd Alloy Electrocatalyst. *Mater. Res. Express* **2019**, *7* (1), 015505.
- (35) Szczuka, C.; Eichel, R. A.; Granwehr, J. Gauging the Importance of Structural Parameters for Hyperfine Coupling Constants in Organic Radicals. *RSC Adv.* **2023**, *13* (21), 14565–14574.
- (36) Cao, N.; Chen, Z.; Zang, K.; Xu, J.; Zhong, J.; Luo, J.; Xu, X.; Zheng, G. Doping Strain Induced Bi-Ti³⁺ Pairs for Efficient N₂ Activation and Electrocatalytic Fixation. *Nat. Commun.* **2019**, *10* (1), 1–12.
- (37) Zhao, Y.; Tao, L.; Dang, W.; Wang, L.; Xia, M.; Wang, B.; Liu, M.; Gao, F.; Zhang, J.; Zhao, Y. High-Indexed PtNi Alloy Skin Spiraled on Pd Nanowires for Highly Efficient Oxygen Reduction Reaction Catalysis. *Small* **2019**, *15* (17), 1–9.
- (38) Li, K.; He, F.; Xie, G.; Wang, Y.; Jiao, M.; Tang, H.; Wu, Z. The Influence of Pt Layers in Ag@Pt NL (111) (n = 1 and 2) on Oxygen Reduction Reaction: A Theoretical Study. *J. Electrochem. Soc.* **2016**, *163* (8), F945–F951.
- (39) Hyman, M. P.; Medlin, J. W. Mechanistic Study of the Electrochemical Oxygen Reduction Reaction on Pt(111) Using Density Functional Theory. *J. Phys. Chem. B* **2006**, *110* (31), 15338–15344.
- (40) Li, J.; Liu, G.; Ren, B.; Croiset, E.; Zhang, Y.; Ricardez-Sandoval, L. Mechanistic Study of Site Blocking Catalytic Deactivation through Accelerated Kinetic Monte Carlo. *J. Catal.* **2019**, *378*, 176–183.
- (41) Stamatakis, M.; Vlachos, D. G. A Graph-Theoretical Kinetic Monte Carlo Framework for on-Lattice Chemical Kinetics. *J. Chem. Phys.* **2011**, *134* (21), 214115.
- (42) Huang, Y.; Nielsen, R. J.; Goddard, W. A.; Soriaga, M. P. The Reaction Mechanism with Free Energy Barriers for Electrochemical Dihydrogen Evolution on MoS₂. *J. Am. Chem. Soc.* **2015**, *137* (20), 6692–6698.
- (43) Rostamikia, G.; Maheshwari, S.; Janik, M. J. Elementary Kinetics of Nitrogen Electroreduction to Ammonia on Late Transition Metals. *Catal. Sci. Technol.* **2019**, *9* (1), 174–181.
- (44) Yu, E.; Pan, Y. Exploring the Hydrogen Evolution Catalytic Activity of the Orthorhombic and Hexagonal Borophene as the Hydrogen Storage Material. *Electrochim. Acta* **2022**, *435* (October), No. 141391.
- (45) Wu, J.; Fan, J.; Zhao, X.; Wang, Y.; Wang, D.; Liu, H.; Gu, L.; Zhang, Q.; Zheng, L.; Singh, D. J.; Cui, X.; Zheng, W. Atomically Dispersed MoO_x on Rhodium Metallene Boosts Electrocatalyzed Alkaline Hydrogen Evolution. *Angew. Chemie - Int. Ed.* **2022**, e202207512.
- (46) Deimel, M.; Prats, H.; Seibt, M.; Reuter, K.; Andersen, M. Selectivity Trends and Role of Adsorbate-Adsorbate Interactions in CO Hydrogenation on Rhodium Catalysts. *ACS Catal.* **2022**, *12* (13), 7907–7917.
- (47) Kristinsdóttir, L.; Skúlason, E. A Systematic DFT Study of Hydrogen Diffusion on Transition Metal Surfaces. *Surf. Sci.* **2012**, *606* (17–18), 1400–1404.
- (48) Zuo, Z.; Liu, S.; Wang, Z.; Liu, C.; Huang, W.; Huang, J.; Liu, P. Dry Reforming of Methane on Single-Site Ni/MgO Catalysts: Importance of Site Confinement. *ACS Catal.* **2018**, *8* (10), 9821–9835.
- (49) Farkas, A.; Hess, F.; Over, H. Experiment-Based Kinetic Monte Carlo Simulations: CO Oxidation over RuO₂ (110). *J. Phys. Chem. C* **2012**, *2* (110), 581–591.
- (50) Campbell, J. M.; Campbell, C. T. The Interactions of Cyclopentane with Clean and Bismuth-Covered Pt(111). *Surf. Sci.* **1989**, *210* (1–2), 46–68.
- (51) Skúlason, E.; Karlberg, G. S.; Rossmeisl, J.; Bligaard, T.; Greeley, J.; Jónsson, H.; Nørskov, J. K. Density Functional Theory Calculations for the Hydrogen Evolution Reaction in an Electrochemical Double Layer on the Pt(111) Electrode. *Phys. Chem. Chem. Phys.* **2007**, *9* (25), 3241–3250.
- (52) Strmcnik, D.; Tripkovic, D.; van der Vliet, D.; Stamenkovic, V.; Marković, N. M. Adsorption of Hydrogen on Pt(1 1 1) and Pt(1 0 0) Surfaces and Its Role in the HOR. *Electrochem. commun.* **2008**, *10* (10), 1602–1605.
- (53) Han, Q.; Luo, Y.; Liu, G.; Wang, Y.; Li, J.; Chen, Z. Comparative Study on the Distinct Activity for NiFe-Based Phosphide and Sulfide Pre-Electrocatalysts towards Hydrogen Evolution Reaction. *J. Catal.* **2022**, *413*, 425–433.
- (54) Xue, Y.; Hui, L.; Yu, H.; Liu, Y.; Fang, Y.; Huang, B.; Zhao, Y.; Li, Z.; Li, Y. Rationally Engineered Active Sites for Efficient and Durable Hydrogen Generation. *Nat. Commun.* **2019**, *10* (1), 2281.

- (55) Ruban, A.; Hammer, B.; Stoltze, P.; Skriver, H.; Nørskov, J. Surface Electronic Structure and Reactivity of Transition and Noble Metals. *J. Mol. Catal. A Chem.* **1997**, *115* (3), 421–429.
- (56) Christmann, K. Interaction of Hydrogen With Solid Surfaces. *Surf. Sci. Rep.* **1988**, *9*, 1–163.
- (57) Le, J.-B.; Chen, A.; Li, L.; Xiong, J.-F.; Lan, J.; Liu, Y.-P.; Iannuzzi, M.; Cheng, J. Modeling Electrified Pt(111)-H_{ad}/Water Interfaces from Ab Initio Molecular Dynamics. *JACS Au* **2021**, *1* (5), 569–577.
- (58) Liu, X.; Jiao, Y.; Zheng, Y.; Jaroniec, M.; Qiao, S. Z. Building Up a Picture of the Electrocatalytic Nitrogen Reduction Activity of Transition Metal Single-Atom Catalysts. *J. Am. Chem. Soc.* **2019**, *141* (24), 9664–9672.
- (59) Xia, B. Y.; Wang, J. N.; Wang, X. X. Synthesis and Application of Pt Nanocrystals with Controlled Crystallographic Planes. *J. Phys. Chem. C* **2009**, *113* (42), 18115–18120.
- (60) Marković, N. M.; Grgur, B. N.; Ross, P. N. Temperature-Dependent Hydrogen Electrochemistry on Platinum Low-Index Single-Crystal Surfaces in Acid Solutions. *J. Phys. Chem. B* **1997**, *101* (27), 5405–5413.
- (61) Gómez, R.; Fernández-Vega, A.; Feliu, J. M.; Aldaz, A. Hydrogen Evolution on Pt Single Crystal Surfaces. Effects of Irreversibly Adsorbed Bismuth and Antimony on Hydrogen Adsorption and Evolution on Pt(100). *J. Phys. Chem.* **1993**, *97* (18), 4769–4776.
- (62) Zhu, S.; Qin, X.; Xiao, F.; Yang, S.; Xu, Y.; Tan, Z.; Li, J.; Yan, J.; Chen, Q.; Chen, M.; Shao, M. The Role of Ruthenium in Improving the Kinetics of Hydrogen Oxidation and Evolution Reactions of Platinum. *Nat. Catal.* **2021**, *4* (8), 711–718.
- (63) Abghoui, Y. Superiority of the (100) Over the (111) Facets of the Nitrides for Hydrogen Evolution Reaction. *Top. Catal.* **2022**, *65* (1–4), 262–269.
- (64) Zhu, S.; Qin, X.; Yao, Y.; Shao, M. PH-Dependent Hydrogen and Water Binding Energies on Platinum Surfaces as Directly Probed through Surface-Enhanced Infrared Absorption Spectroscopy. *J. Am. Chem. Soc.* **2020**, *142* (19), 8748–8754.
- (65) Osawa, M.; Tsushima, M.; Mogami, H.; Samjeské, G.; Yamakata, A. Structure of Water at the Electrified Platinum-Water Interface: A Study by Surface-Enhanced Infrared Absorption Spectroscopy. *J. Phys. Chem. C* **2008**, *112* (11), 4248–4256.



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